A13483W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4: Sub-Atomic Physics

TRINITY TERM 2015

Friday, 19 June, 2.30 pm – 4.30 pm

10 minutes reading time

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) The rate of alpha decay of a nucleus can be expressed in terms of the probability of finding an alpha particle in the nucleus, the frequency of attempts for an alpha particle to escape, and the tunneling probability for the alpha particle to escape. Calculate the tunneling probability for a one-dimensional Hamiltonian

$$H = p^2/2m + V \ ,$$

where all symbols have their usual meaning, $V = 2Z\alpha_{\rm EM}/r$, $\alpha_{\rm EM}$ is the fine structure constant, and r is the distance from the centre of the nucleus. Express your answer in terms of the energy released in the decay (Q), the radius of the nucleus ($r_{\rm a}$) and the atomic number of the daughter nucleus (Z). [9]

(b) The alpha decay of ${}^{244}_{98}$Cf has a Q-value of 7.33 MeV and a half-life of 19.4 minutes. Calculate the half-life of ${}^{224}_{90}$Th given its Q-value of 5.52 MeV and assuming it has the same probability and frequency of finding alpha particles in the nucleus as ${}^{244}_{98}$Cf. Assume the nuclear radius of ${}^{244}_{98}$Cf is 9.44 fm and the nuclear radius of ${}^{224}_{90}$Th is 9.27 fm. [7]

(c) Calculate the binding energy B of the alpha decay product of ${}^{224}_{90}$Th using the formula

$$B = \alpha A - \beta A^{2/3} - \gamma \, \frac{(A-2Z)^2}{A} - \epsilon \, \frac{Z^2}{A^{1/3}} + \delta \ ,$$

where A is the mass number, and the parameters $\alpha = 16$, $\beta = 18$, $\gamma = 23$, $\epsilon = 0.7$, and $\delta = 11/\sqrt{A}$ (in units of MeV). State what each term represents. Assuming the alpha decay dominates, is the ${}^{224}_{90}$Th decay product likely to have a shorter or longer half-life than ${}^{224}_{90}$Th? [9]

2. (a) For each of the following decays, state the interaction by which it proceeds, and give an example quark-flow or tree-level Feynman diagram (or say why the decay is not allowed):

(i) $\mathrm{p} \to \pi^+ + \pi^0 + \pi^0$
(ii) $\mathrm{n} \to \mathrm{p} + \mathrm{e}^- + \bar{\nu}_{\rm e}$
(iii) $\pi^0 \to \gamma + \gamma$
(iv) $\mathrm{K}^0 \to \pi^+ + \pi^-$
(v) $D^+ \to \mathrm{K}^0 + \pi^+$
(vi) $\Delta^+ \to \mathrm{p} + \pi^0$ [6]

(b) State which decay you expect to have the higher branching ratio, $\Xi^- \to \Lambda + \pi^-$ or $\Xi^- \to \mathrm{n} + \pi^-$, and why. The Ξ^- baryon has strangeness quantum number -2. [2]

(c) The J/ψ meson can decay via a virtual photon. If this were the only possible decay mode, what would be the branching fraction for $J/\psi \to \mu^+ + \mu^-$? [7]

(d) The J/ψ meson can also decay via a three-gluon intermediate state. Using the observed branching ratio of 6% for $J/\psi \to \mu^+ + \mu^-$, estimate the value of the strong coupling constant $\alpha_{\rm s}$ and state any assumptions you make. [10]

3. (a) Assume a nucleus of charge Z and radius a has a uniform charge distribution. Calculate the electromagnetic form factor $F(\Delta k)$ associated with this nucleus, where Δk is the change in momentum of a projectile. [10]

(b) Determine the differential cross section $\mathrm{d}\sigma/\mathrm{d}\Omega$ for an electron scattering off this nucleus, using the Born approximation and the Rutherford cross section,

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{Rutherford}} = \frac{Z^2\,\alpha_{\text{EM}}^2}{4\,E^2\sin^4(\theta/2)}\ ,$$ [4]

where α_{EM} is the fine structure constant, E is the energy of the incident electron, and θ is the scattering angle.

(c) An electron with energy 1 GeV scatters off a proton at rest in the laboratory. Calculate the differential cross section $\mathrm{d}\sigma/\mathrm{d}\Omega$ for an observed electron with energy 800 MeV. Assume that the proton charge distribution is uniform within a radius of 1 fm. [6]

(d) How realistic is the assumption of a uniform charge distribution for a proton at high scattering energies? What interactions within the proton affect the charge distribution as the scattering energy increases? At very high energies what particles within the proton do the electrons scatter off? [5]

4. (a) The W boson mass has been measured to high precision at electron-positron and proton-antiproton colliders using the processes $\mathrm{e}^+ + \mathrm{e}^- \to W^+ + W^-$ and $\mathrm{p} + \bar{\mathrm{p}} \to W + X$, respectively. Draw two lowest-order Feynman diagrams for the $\mathrm{e}^+ + \mathrm{e}^- \to W^+ + W^-$ process and one for the $\mathrm{p} + \bar{\mathrm{p}} \to W + X$ process. Why is it not possible to produce a single W boson with no additional particle in an electron-positron collider? [5]

(b) Write down the decay modes of the W boson. What are the best decay modes for observing the W boson at a proton-antiproton collider? What is the branching fraction for each of these best decay modes? [5]

(c) The cross section for W boson production at a proton-antiproton collider with a centre-of-mass energy of 1.96 TeV is 25 nb. The total cross section for inelastic scattering is 30 mb, and 3×10^{14} inelastic events were produced in each detector by the Tevatron. How many W boson events were produced in each detector? How many W bosons were produced in the best decay modes for observation in each detector [from (b)]? [3]

(d) In proton-antiproton collisions with a centre-of-mass energy of 2 TeV, a W boson is produced by constituents of the proton and the antiproton. If these interacting particles have a fraction x_{p} of the proton's energy and $x_{\bar{\mathrm{p}}}$ of the anti-proton's energy, what is the value of the product $x_{\mathrm{p}} x_{\bar{\mathrm{p}}}$ if they produce a single W boson with mass equal to 80 GeV? In the case where the fractions are equal, what is $x = x_{\mathrm{p}} = x_{\bar{\mathrm{p}}}$? [3]

(e) The mass of the W boson produced can be reconstructed as the centre-of-mass energy of the final-state $l\nu$ system. Writing the four-vectors of the final-state particles as $(E^l, p_x^l, p_y^l, p_z^l)$ and $(E^\nu, p_x^\nu, p_y^\nu, p_z^\nu)$, write down the square of the centre-of-mass energy of this system (neglecting lepton mass) and use it to obtain the expression for m, the invariant mass of the system. [3]

(f) How is the neutrino momentum measured in a collider experiment? Why is it not possible to measure the neutrino momentum along the beam direction in a proton-antiproton collider? [3]

(g) In the expression for the invariant mass, set p_z^l and p_z^ν to zero to obtain an expression for the transverse mass m_{T}. [2]

(h) In the case where a charged lepton and neutrino have opposite non-zero p_z, is the transverse mass larger or smaller than the invariant mass? [1]